Question 1

Display detailed file permissions of a selected directory

Approach

I selected a directory and used the long listing option to view detailed file information, including read, write, and execute permissions.

Commands

- `ls` → lists directory contents
- `ls -l` → shows detailed file permissions
- `ls -la` → includes hidden files

Output

STEP 1 : Navigate to directory

STEP 2 : Display long listing

```
┌──(aman㉿kali)-[~]
└─$ ls -l
total 56
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Desktop
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Documents
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Downloads
-rw-rw-r--  1 aman aman    39 Mar 22 12:01 file.txt
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Music
drwxrwxr-x  2 aman aman  4096 Mar 22 12:12 newfile
drwxrwxr-x 26 aman aman 12288 Mar  8 13:09 nmap
drwxr-xr-x  2 aman aman  4096 Mar 28 21:29 Pictures
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Public
drwxrwxr-x 11 aman aman  4096 Mar 28 21:56 sqlmap
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Templates
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Videos
```

STEP 3 : Display hidden files

```
┌──(aman㉿kali)-[~]
└─$ ls -la
total 208
drwx------ 21 aman aman  4096 Mar 28 21:54 .
drwxr-xr-x  3 root root  4096 Mar  8 10:46 ..
-rw-r--r--  1 aman aman   220 Mar  8 10:46 .bash_logout
-rw-r--r--  1 aman aman  5551 Mar  8 10:46 .bashrc
-rw-r--r--  1 aman aman  3526 Mar  8 10:46 .bashrc.original
drwxrwxr-x 10 aman aman  4096 Mar 28 21:53 .cache
drwxr-xr-x 14 aman aman  4096 Mar 10 04:28 .config
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Desktop
-rw-r--r--  1 aman aman    35 Mar  8 10:52 .dmrc
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Documents
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Downloads
-rw-r--r--  1 aman aman 11759 Mar  8 10:46 .face
lrwxrwxrwx  1 aman aman     5 Mar  8 10:46 .face.icon → .face
-rw-rw-r--  1 aman aman    39 Mar 22 12:01 file.txt
drwx------  3 aman aman  4096 Mar  8 10:52 .gnupg
-rw-------  1 aman aman     0 Mar  8 10:52 .ICEauthority
drwxr-xr-x  3 aman aman  4096 Mar  8 10:46 .java
drwxr-xr-x  5 aman aman  4096 Mar  8 10:52 .local
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Music
drwxrwxr-x  2 aman aman  4096 Mar 22 12:12 newfile
drwxrwxr-x 26 aman aman 12288 Mar  8 13:09 nmap
drwxr-xr-x  2 aman aman  4096 Mar 28 21:29 Pictures
-rw-r--r--  1 aman aman   807 Mar  8 10:46 .profile
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Public
drwxrwxr-x 11 aman aman  4096 Mar 28 21:56 sqlmap
drwx------  3 aman aman  4096 Mar  8 10:52 .ssh
drwxrwxr-x  3 aman aman  4096 Mar  8 12:34 .subversion
-rw-r--r--  1 aman aman     0 Mar  8 11:32 .sudo_as_admin_successful
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Templates
-rw-r-----  1 aman aman     5 Mar 28 19:11 .vboxclient-clipboard-tty7-control.pid
-rw-r-----  1 aman aman     5 Mar 28 19:11 .vboxclient-clipboard-tty7-service.pid
-rw-r-----  1 aman aman     5 Mar 28 19:11 .vboxclient-draganddrop-tty7-control.pid
-rw-r-----  1 aman aman     5 Mar 28 19:11 .vboxclient-draganddrop-tty7-service.pid
-rw-r-----  1 aman aman     5 Mar 28 19:11 .vboxclient-hostversion-tty7-control.pid
-rw-r-----  1 aman aman     5 Mar 28 19:11 .vboxclient-seamless-tty7-control.pid
-rw-r-----  1 aman aman     5 Mar 28 19:11 .vboxclient-seamless-tty7-service.pid
-rw-r-----  1 aman aman     5 Mar 28 19:11 .vboxclient-vmsvga-session-tty7-control.pid
-rw-r-----  1 aman aman     5 Mar 28 19:11 .vboxclient-vmsvga-session-tty7-service.pid
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Videos
-rw-------  1 aman aman    49 Mar 28 19:11 .Xauthority
-rw-------  1 aman aman  4472 Mar 28 21:29 .xsession-errors
-rw-------  1 aman aman  4406 Mar 24 21:29 .xsession-errors.old
drwxrwxr-x  2 aman aman  4096 Mar  8 13:26 .zenmap
-rw-r--r--  1 aman aman   336 Mar  8 10:46 .zprofile
-rw-------  1 aman aman  2260 Mar 24 06:16 .zsh_history
-rw-r--r--  1 aman aman 10855 Mar  8 10:46 .zshrc
```

Key Observations

Each file shows permissions in this format : `-rwxr-xr-x`

What I learned

Linux controls access using file permissions which define who can read. modify, or execute files.

Question 2

Grant execute permission to a file using both numeric and symbolic methods

Approach

I selected a file without execute permission and modified permissions using the `chmod` command. I applied both, numeric (777) and symbolic (u+x) permission methods.

Commands

- `chmod` → change file permissions
- numeric method → uses numbers
- symbolic method → uses letters
  Permission values :
- 4 → read
- 2 → write
- 1 → execute

Output

STEP 1 : Create test file

STEP 2 : Check permissions

```
┌──(aman㉿kali)-[~]
└─$ ls -l
total 56
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Desktop
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Documents
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Downloads
-rw-rw-r--  1 aman aman    39 Mar 22 12:01 file.txt
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Music
drwxrwxr-x  2 aman aman  4096 Mar 22 12:12 newfile
drwxrwxr-x 26 aman aman 12288 Mar  8 13:09 nmap
drwxr-xr-x  2 aman aman  4096 Mar 28 21:29 Pictures
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Public
drwxrwxr-x 11 aman aman  4096 Mar 28 21:56 sqlmap
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Templates
-rw-rw-r--  1 aman aman     0 Mar 29 01:48 testfile.sh
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Videos
```

STEP 3 : Give execute permission using numeric method

```
┌──(aman㉿kali)-[~]
└─$ chmod 777 testfile.sh
```

STEP 4 : Give execute permission using symbolic method

```
┌──(aman㉿kali)-[~]
└─$ chmod u+x testfile.sh
```

STEP 5 : Verify permission change

```
┌──(aman㉿kali)-[~]
└─$ ls -l
total 56
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Desktop
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Documents
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Downloads
-rw-rw-r--  1 aman aman    39 Mar 22 12:01 file.txt
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Music
drwxrwxr-x  2 aman aman  4096 Mar 22 12:12 newfile
drwxrwxr-x 26 aman aman 12288 Mar  8 13:09 nmap
drwxr-xr-x  2 aman aman  4096 Mar 28 21:29 Pictures
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Public
drwxrwxr-x 11 aman aman  4096 Mar 28 21:56 sqlmap
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Templates
-rwxrwxrwx  1 aman aman     0 Mar 29 01:48 testfile.sh
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Videos
```

Key Observation

Permission changes reflect immediately in the file listing

`-rwxrwxrwx`

What I learned

Permissions can be modified numerically or symbolically to control file access.

Question 3

Change file ownership using chown command

Approach

I selected a file and changed its owner using the chown command.

Commands

- `chown` → change file ownership
- syntax : `sudo chown user:group filename`

Output

STEP 1 : Create new file

```
┌──(aman㉿kali)-[~]
└─$ touch ownershipfile.txt
```

STEP 2 : Change ownership

```
┌──(aman㉿kali)-[~]
└─$ sudo chown root:root ownershipfile.txt
[sudo] password for aman:
```

STEP 3 : Verify ownership change

```
┌──(aman㉿kali)-[~]
└─$ ls -l
total 52
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Desktop
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Documents
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Downloads
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Music
drwxrwxr-x  2 aman aman  4096 Mar 22 12:12 newfile
drwxrwxr-x 26 aman aman 12288 Mar  8 13:09 nmap
-rw-rw-r--  1 root root     0 Mar 29 02:00 ownershipfile.txt
drwxr-xr-x  2 aman aman  4096 Mar 28 21:29 Pictures
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Public
drwxrwxr-x 11 aman aman  4096 Mar 28 21:56 sqlmap
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Templates
drwxr-xr-x  2 aman aman  4096 Mar  8 10:52 Videos
```

Key Observations

Ownership is displayed in the file listing :
`owner group filename`

What I learned

Ownership determines which users have control over files and directories.

Question 4

Find files that have the SGID permission bit enabled

Approach

I used the find command to search the system for files with the SGID bit set.
SGID allows files to run with group privileges.

Commands

- `find` → search for files
- SGID value → 2000 permission bit

Output

STEP 1 : Search for SGID files

```
┌──(aman㉿kali)-[~]
└─$ find / -perm -2000 2>/dev/null
/run/tpm2-tss/eventlog
/run/postgresql
/run/log/journal
/var/lib/tpm2-tss/system/keystore
/var/mail
/var/log/mysql
/var/log/journal
/var/log/journal/f039f01c084e49ffae785f7936d5ce0b
/var/log/redis
/usr/lib/xorg/Xorg.wrap
/usr/bin/plocate
/usr/bin/chage
/usr/bin/ssh-agent
/usr/bin/crontab
/usr/bin/expiry
/usr/bin/dotlockfile
/usr/local/lib/python2.7
/usr/local/lib/python2.7/dist-packages
/usr/local/lib/python2.7/site-packages
/usr/sbin/unix_chkpwd
/etc/chatscripts
/etc/redis
/etc/ppp/peers
```

Explanation :

- `/` → search entire system
- `-perm -2000` → find SGID files
- `2>/dev/null` → hides permission errors

Key Observations

SGID files allow execution with group permissions rather than user permissions.

What I learned

Special permission bits like SGID can affect privilege levels and are important in security auditing.

Security Insights

- Viewing directory permissions helps identify files that attackers may access without authorization.
- Incorrect execute permissions may allow malicious scripts to run on a system.
- Incorrect ownership can allow unauthorized modifications

- Improper file ownership can allow privilege abuse or unauthorized modification of critical files.
- SGID files can be exploited to execute programs with elevated group privileges.